Punnett Squares

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Punnett squares (named after R. C. Punnett, an English geneticist, 1875–1967) are used in genetics to calculate the frequencies of the different genotypes and phenotypes among the offspring of a cross. Punnett used poultry and sweet peas to discovered many genetic linkages.

1 Give a brief definition of 'genotypes'. _____

2 Give a brief definition of 'phenotypes'. _____

We represent the dominant of the two alleles with an upper-case letter and the recessive allele with a lower-case letter. The male (♂) parent is shown on the upper part of the Punnett square and the female (♀) parent is shown on the side.

	T	T
t	Tt	Tt
t	Tt	Tt

TT is the genotype for homozygous tall plants and tt is the genotype for homozygous small plants. We are going to see what happens if we cross these.

A 'Punnett square' can be used, which shows that all of the first generation offspring (F_1) will be heterozygous (Tt) plants. Because tall is dominant (in this case) over short, all of these plants will be tall.

	T	t
T	TT	Tt
t	Tt	tt

If these offspring plants are then crossed, the next generation (F_2) will be TT, Tt and tt, in the ratio of 1 : 2 : 1. One quarter will be homozygous tall (TT), one half will be heterozygous tall (Tt) and one quarter will be homozygous small (tt).

A male German Shepherd with erect ears (EE) was mated with a female German Shepherd with floppy ears (ee).

3 Draw a Punnett square in the box provided on the right, showing the possible outcomes of the offspring of this cross.

German Shepherd Cross

4 Complete this summary table for the German Shepherd cross described above:

Offspring	Genotype	Phenotype
a		
b		
c		
d		

5 Summarise the rules for representing dominant and recessive alleles in Punnett squares.

_____

6 What is meant by 'genetic linkage'? _____

_____

7 What is meant by the 'F_2 generation'? _____

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